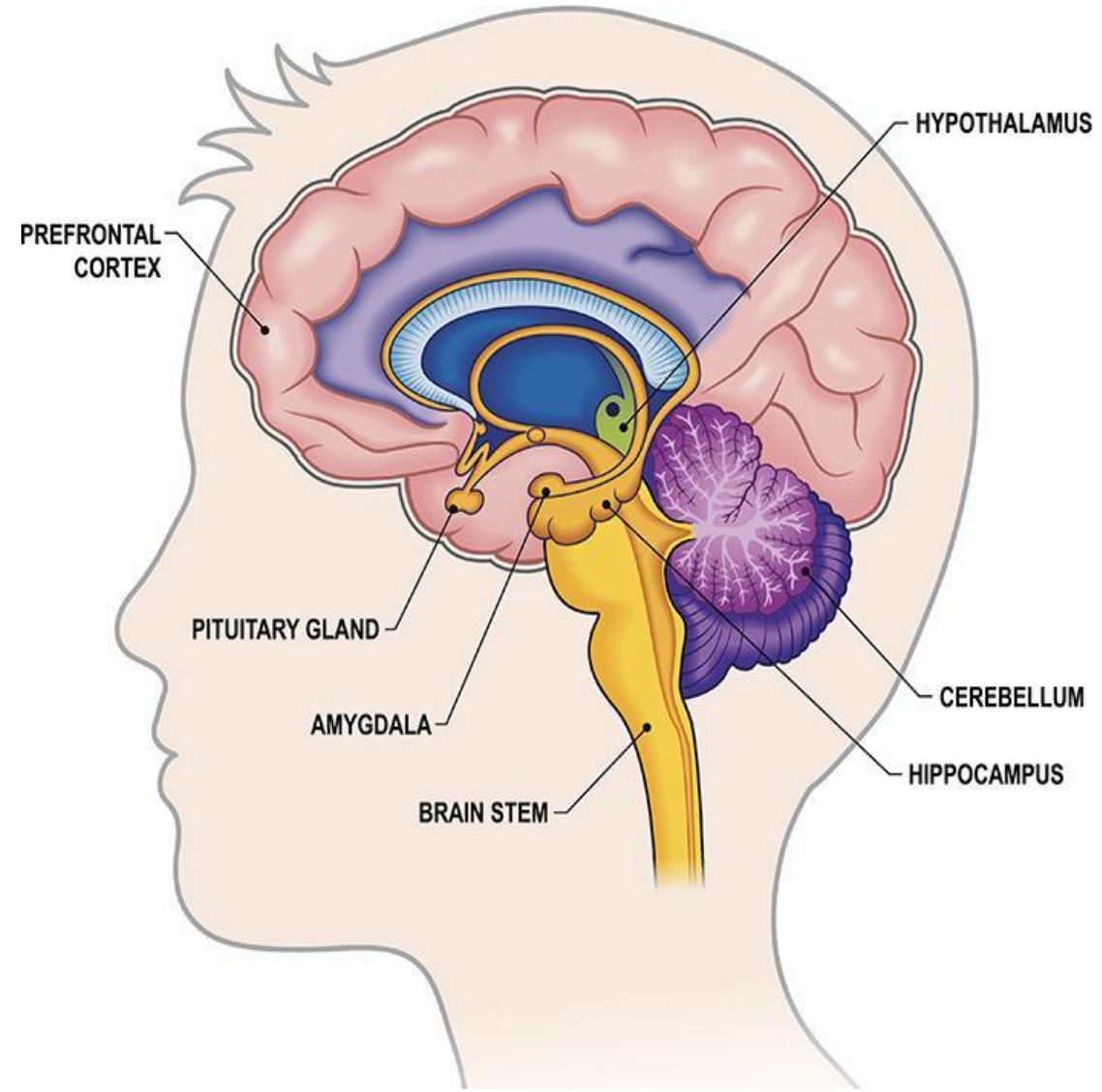


LIMBIC SYSTEM

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LEARNING OBJECTIVES:

- **Describe the anatomical components and major subdivisions** of the limbic system, including cortical and subcortical structures, and their interconnections.
- **Explain the functional roles** of the limbic system in emotional and behavioral regulation.
- **Analyze the contributions of key structures** such as the hypothalamus, hippocampus, and amygdala in mediating physiological and behavioral responses.
- **Describe the clinical relevance** of limbic system dysfunction, including syndromes such as Klüver-Bucy, Korsakoff, and Alzheimer's disease.

LIMBIC SYSTEM:

"Limbic" = Border (forms a ring around the brainstem and corpus callosum).

- Referred to as the **"Emotional Nervous System."** because of its role in processing emotions and linking them to behavioral responses.
- It plays a key role in emotional responses, forming memories, and linking emotions to sensory experiences.

- **Primary Function:**

Controls **emotional behavior, motivational drives, and learning/memory consolidation.**

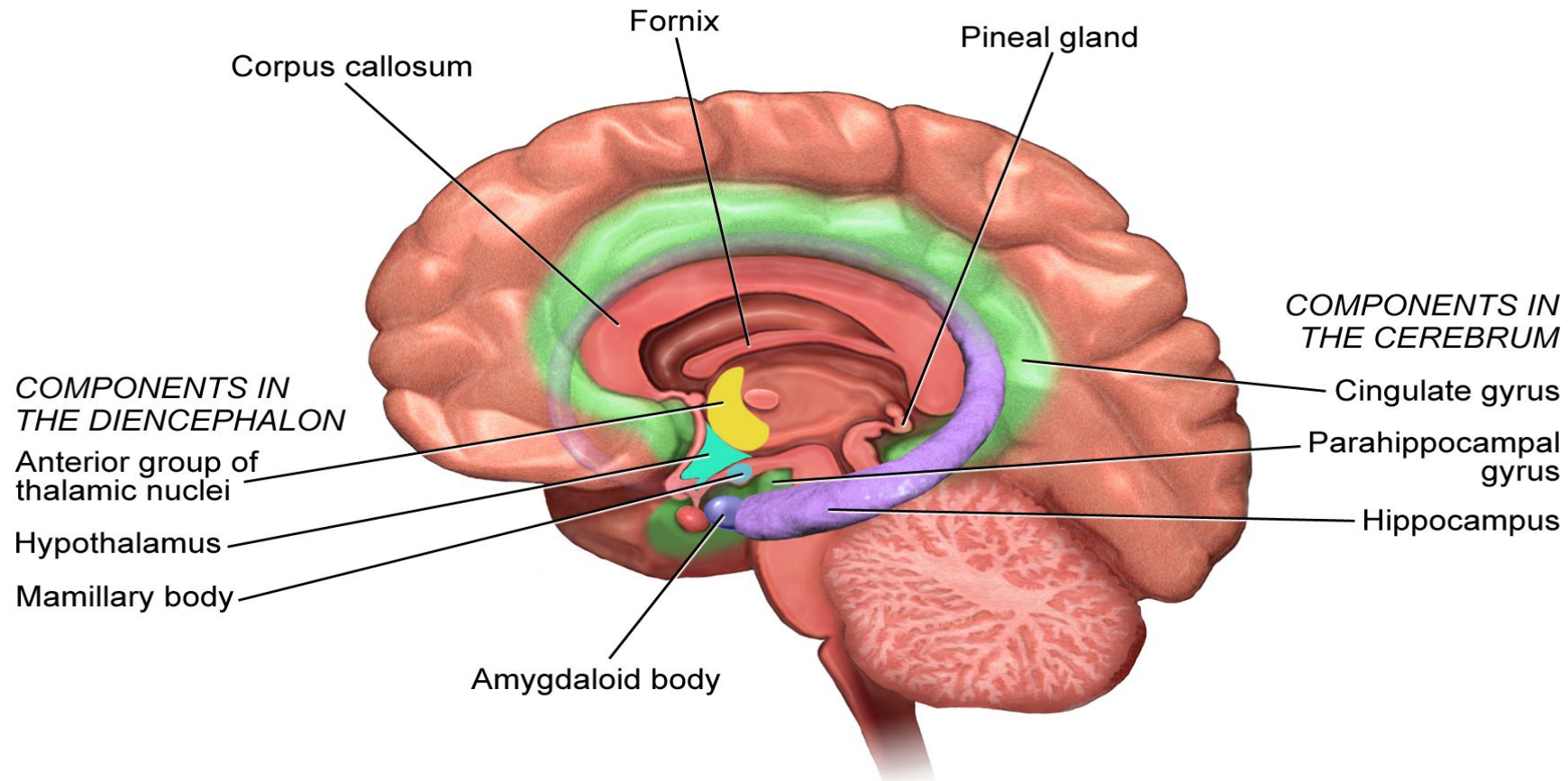
- **FUNCTIONAL ANATOMY OF LIMBIC SYSTEM:**

The limbic system includes various subcortical structures such as hypothalamus, septum, parolfactory area, anterior nucleus of thalamus, amygdala, hippocampus, portions of basal ganglia.

- **Brainstem Connection:**

Directly linked to the brainstem reticular formation via the **medial forebrain bundle.**

The Limbic System

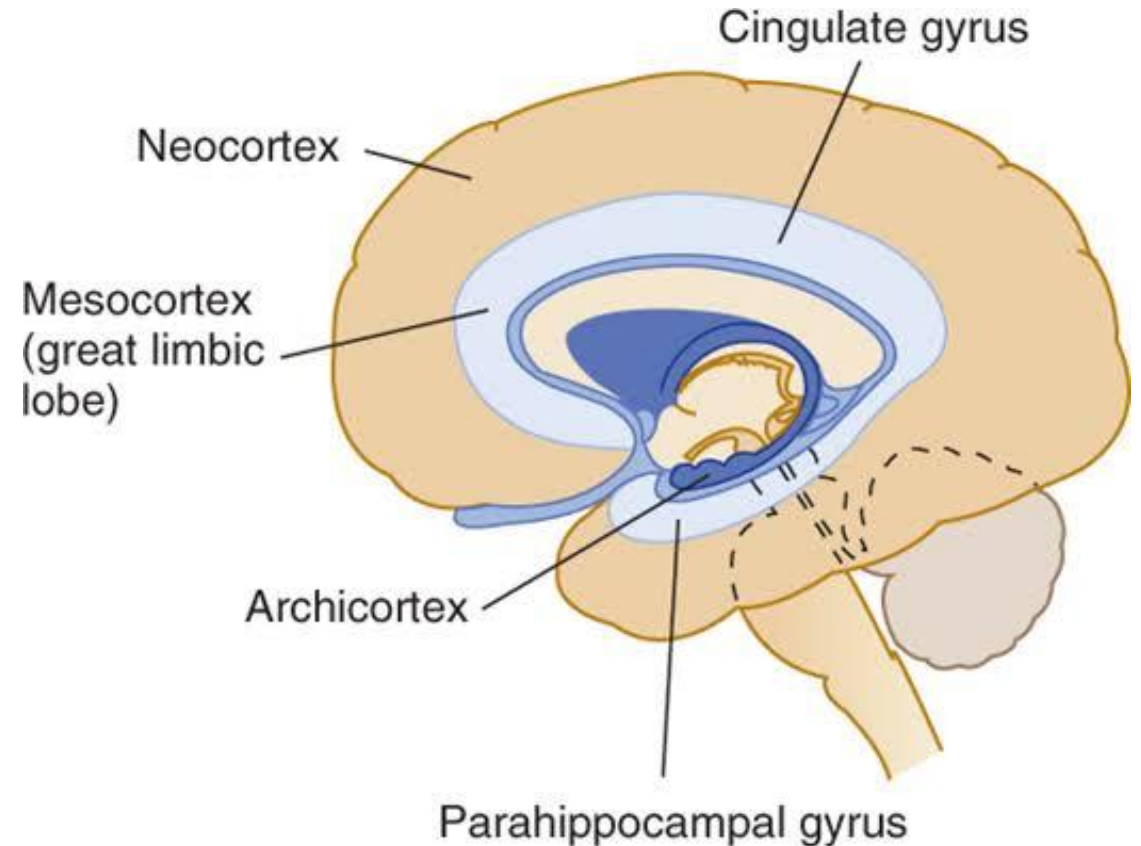


The Limbic System includes components from both the diencephalon (anterior group of thalamic nuclei, hypothalamus, and mammillary body) and the cerebrum (cingulate gyrus, parahippocampal gyrus, hippocampus, and amygdaloid body). Structures additionally labeled for reference are the corpus callosum, fornix, and pineal gland.

COMPONENTS OF LIMBIC SYSTEM

Structures of limbic system are classified into four groups:

1. Archicortical structures
2. Paleocortical structures
3. Juxtallo cortical structures
4. Subcortical structures



COMPONENTS OF LIMBIC SYSTEM:

ARCHICORTICAL STRUCTURES

- **Archicortex** forms **allocortex** along with **paleocortex** . Archicortex is the phylogenetically oldest structure. It is concerned with **memory**.

PALEOCORTICAL STRUCTURES

- Paleocortex is in between archicortex and neocortex.
- It is concerned with **olfaction**.

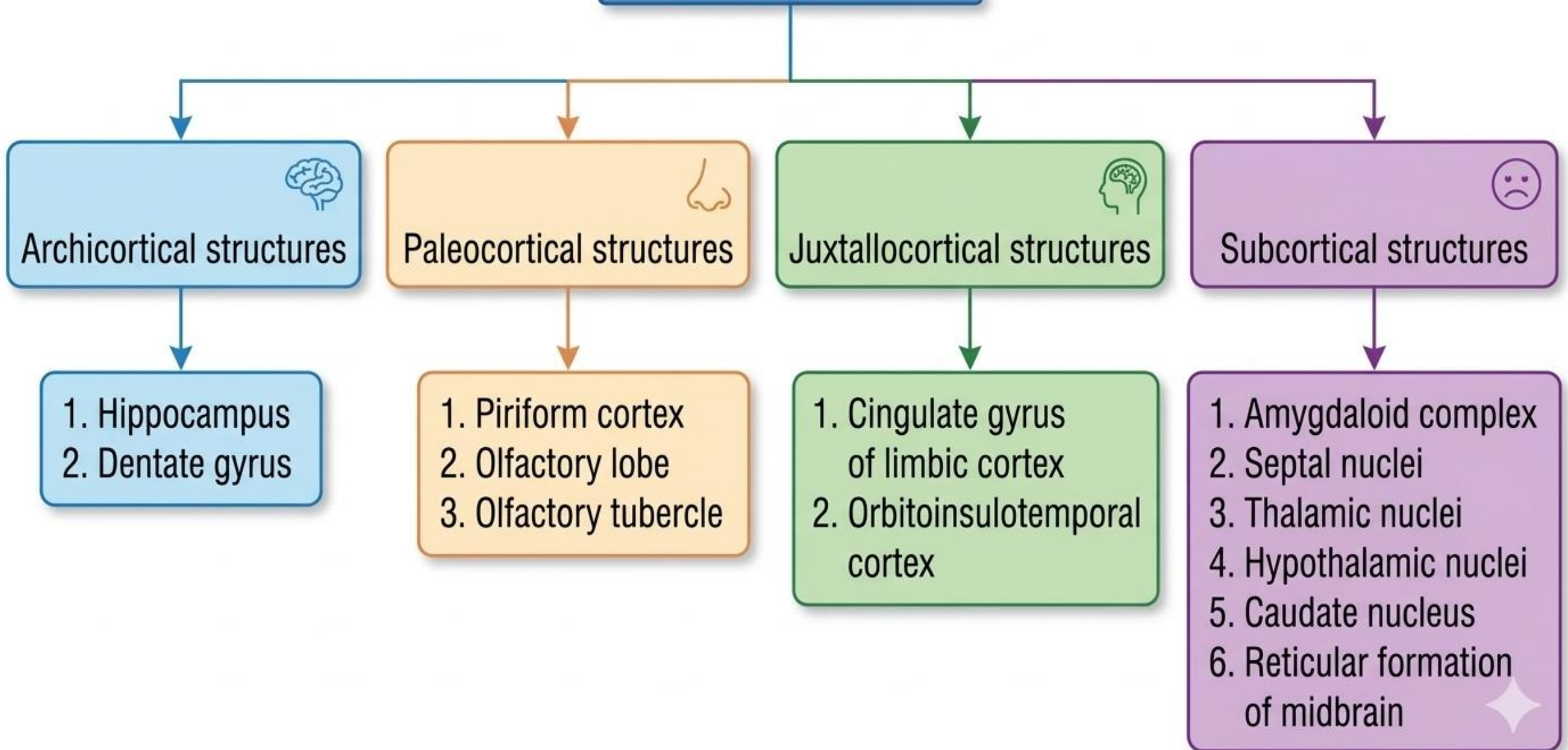
JUXTALLOCORTICAL STRUCTURES

- Juxtallocortex or **mesocortex** is situated between pale cortex and neocortex.

SUBCORTICAL STRUCTURES

- Structures situated below the level of cortex are called subcortical structures. Limbic system includes six subcortical structures.

Limbic system



CONNECTIONS OF LIMBIC SYSTEM:

Connections of limbic system are complex. Following are the major (afferent and efferent) connections of limbic system:

1. **Fornix:** It includes fibers connecting:
 - i. Hippocampus and septal nuclei with the mammillary body
 - ii. Hippocampus with hypothalamic nuclei.
2. **Lateral hypothalamus** receives afferent fibers from:
 - i. Hippocampus
 - ii. Septal nuclei
 - iii. Olfactory tubercle
 - iv. Head of caudate nucleus
 - v. Piriform area
 - vi. Periamygdaloid area

3. **Caudate nucleus** receives fibers from:

- i. Cingulate gyrus
- ii. Intralaminar nuclei of thalamus.

4. **Brainstem reticular formation** receives fibers from:

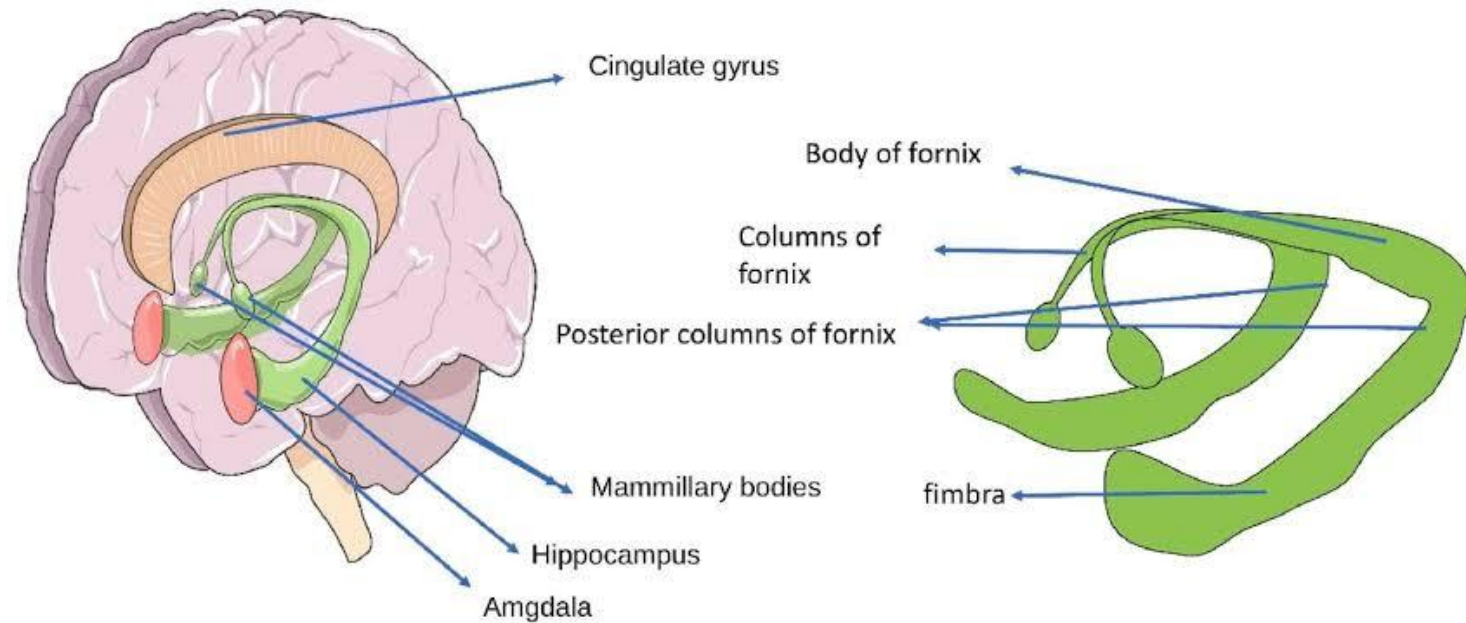
- i. Hippocampus
- ii. Cingulate gyrus.

5. **Papez circuit**

- **The Papez Circuit Sequence** –Papez circuit is the interconnections between various structures of limbic system, which form a complex of closed circuit.
- **Function**: Vital for episodic memory ,and memory encoding of brain.

**A step-by-step description of the classical memory loop
(Hippocampus → Fornix → Mammillary Bodies → Anterior Thalamus
→ Cingulate Gyrus → Parahippocampal Gyrus)**

- **Mammillary Body:** A pair of small, round, bulbous structures located at the base of the brain in the posterior hypothalamus.
- **Cingulate Gyrus:** A large, arch-shaped cortical area situated above the corpus callosum on the inner, medial surface of the brain.



KEY COMPONENT OF LIMBIC SYSTEM :

- **HYPOTHALAMUS:** Regulates basic functions such as hunger, thirst, sleep, and body temperature. It also plays a role in the release of hormones that affect the body's response to stress.
- **AMYGDALA:** Involved in processing emotions like fear and pleasure, and plays a role in emotional memory.
- **HIPPOCAMPUS:** Critical for memory formation and spatial navigation.
- **THALAMUS:** Relays sensory information to other parts of the brain, influencing how emotional experiences are interpreted.
- **CINGULATE GYRUS:** Involved in processing emotions and regulating autonomic functions, such as heart rate and blood pressure.

LIMBIC CORTEX:

- Forms a ring around these deep structures, comprising:
- Orbitofrontal area (Ventral surface of the frontal lobes)
- Subcallosal gyrus
- Cingulate gyrus (medial aspect of the cerebral hemisphere)
- Parahippocampal gyrus and uncus (ventromedial surface of the temporal lobe)
- The limbic cortex serves as communication link between the neocortex (responsible for higher cognitive functions) and the deeper, more primitive limbic structures (regulating emotion and behavior).

ANATOMY OF LIMBIC SYSTEM:

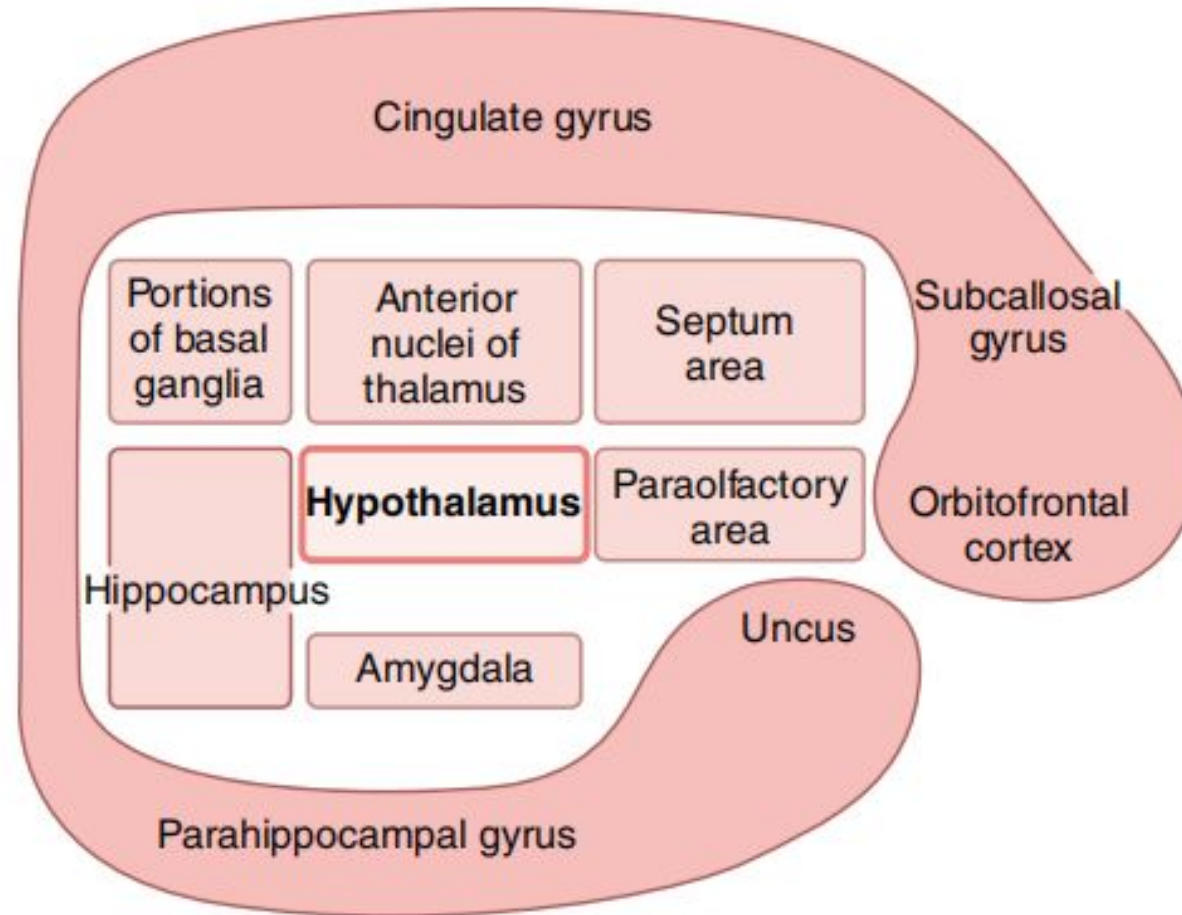


Figure 58-5 Limbic system, showing the key position of the hypothalamus.

HYPOTHALAMUS:

- The hypothalamus is a small but vital control center for the limbic system, makes up less than 1% of brain mass but it is crucial for overall brain and body functions
- It communicate with all levels of the limbic system
- It sends output signals in three directions:
 1. To the brain stems for autonomic control
 2. To higher brain areas (diencephalon and cerebrum) for emotional and cognitive functions
 3. To the pituitary gland to regulate endocrine functions

POSTERIOR

Dorsomedial nucleus
(GI stimulation)

Posterior hypothalamus
(Increased blood pressure)
(Pupillary dilation)
(Shivering)

Perifornical nucleus
(Hunger)
(Increased blood pressure)
(Rage)

Ventromedial nucleus
(Satiety)
(Neuroendocrine control)

Mamillary body
(Feeding reflexes)

Arcuate nucleus and periventricular zone
(Hunger)
(Satiety)
(Neuroendocrine control)

Lateral hypothalamic area (not shown)
(Thirst and hunger)

Hypothalamus

ANTERIOR

Paraventricular nucleus
(Oxytocin release)
(Water conservation)
(Satiety)

Medial preoptic area
(Bladder contraction)
(Decreased heart rate)
(Decreased blood pressure)

Posterior preoptic and
anterior hypothalamic areas
(Body temperature regulation)
(Panting)
(Sweating)
(Thyrotropin inhibition)

Optic chiasm (Optic nerve)

Supraoptic nucleus
(Vasopressin release)

Infundibulum



Hypothalamic Functions, Roles, and Mechanisms

Function	Hypothalamic Area	Role	Mechanisms
Cardiovascular Regulation	Posterior and lateral hypothalamus	↑ Arterial pressure & heart rate	Acts through cardiovascular control centers in the pons and medulla
	Preoptic area	↓ Arterial pressure & heart rate	Acts through cardiovascular control centers in the pons and medulla
Body Temperature Regulation	Preoptic area	Senses blood temperature and regulates heat-loss or conservation mechanisms	• ↑ Temp: activates heat-loss mechanisms • ↓ Temp: activates heat-conservation mechanisms

Body Water Regulation	Lateral hypothalamus	Stimulates thirst when body fluids are concentrated	Drives the sensation of thirst
	Supraoptic nuclei	Releases vasopressin (ADH) via the posterior pituitary	Reduces urine output by increasing water reabsorption in the kidneys
Uterine Contractility & Milk Ejection	Paraventricular nuclei	Secretes oxytocin	Stimulates uterine contractions in labor & milk ejection during breastfeeding
Gastrointestinal & Feeding Regulation	Lateral hypothalamic area	Drives hunger	Damage causes loss of appetite
	Ventromedial nucleus	Satiety center – suppresses hunger	Destruction leads to overeating and obesity
	Mammillary bodies	Coordinates feeding reflexes	Controls reflexes like licking and swallowing

Endocrine Regulation

Hypothalamic-pituitary axis

Controls anterior pituitary secretion via releasing/inhibitory hormones

Regulates growth, reproduction, metabolism, and other endocrine functions

Circadian Regulation

Suprachiasmatic nucleus

Sleep-wake cycles, cognition, mood, appetite metabolism, autonomic nervous system activity

SCN operates through CLOCK & BMAL1 transcription factors which activate PER and CRY genes

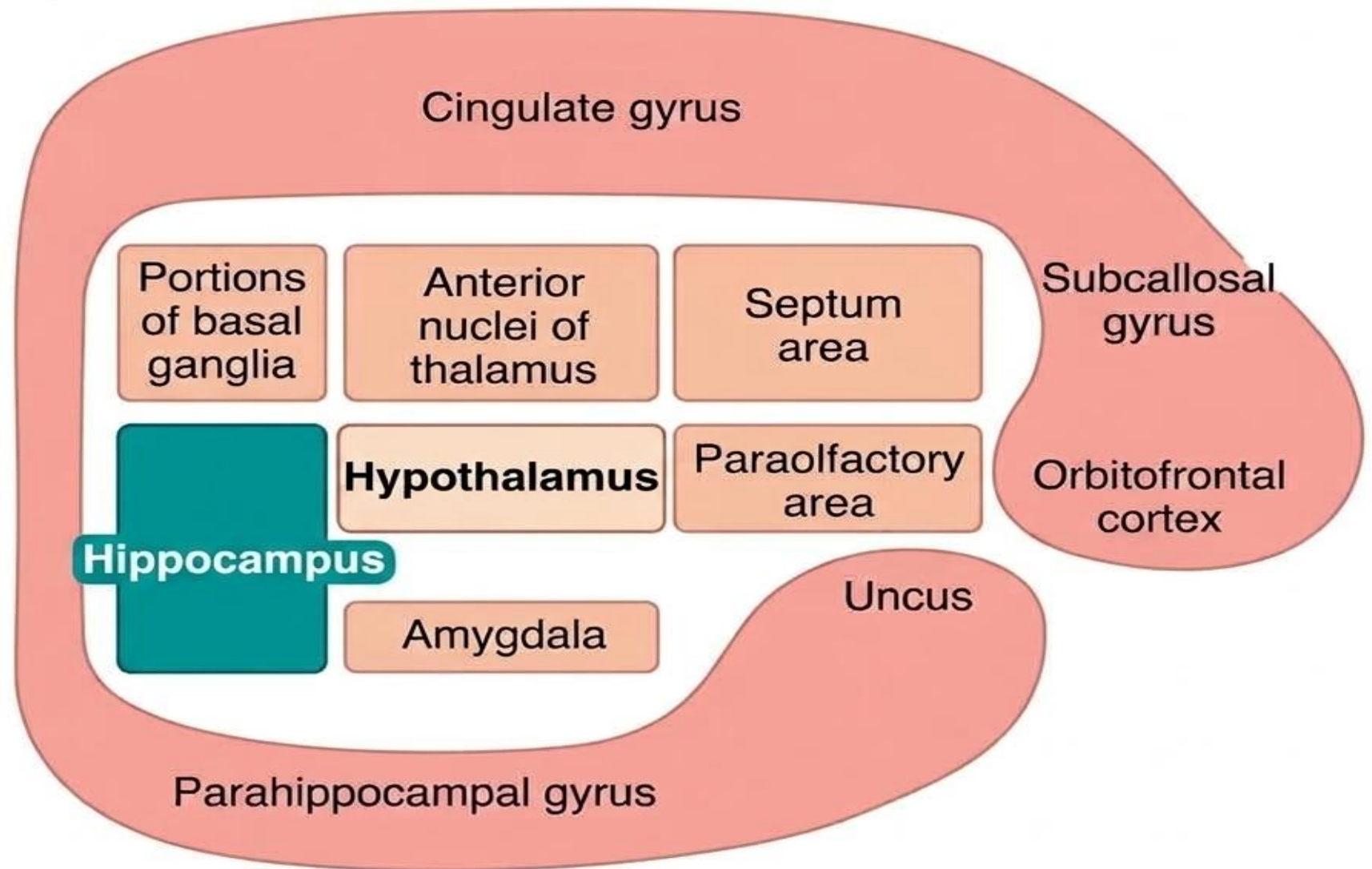
Hypothalamic Behavioral and Emotional Influences

Category / Aspect	Hypothalamic Target / Area	Key Features & Functional Effects
Behavioral Effects of Stimulation	Lateral Hypothalamus	Thirst, eating, increased activity, rage, and aggression.
	Ventromedial Nucleus	Satiety, decreased eating, tranquility (<i>opposite of lateral hypothalamus</i>).
	Periventricular Nuclei	Fear and punishment reactions.
	Anterior and Posterior Hypothalamus	Sexual drive stimulation.

Effects of Hypothalamic Lesions	Lateral Hypothalamus Lesions	Decreased eating/drinking, starvation, passivity, and loss of drives.
	Ventromedial Hypothalamus Lesions	Excessive eating/drinking, hyperactivity, and extreme rage.
Reward Centers	Location	Medial forebrain bundle, lateral & ventromedial hypothalamus, septum, amygdala, thalamus, basal ganglia, mesencephalon.
	Function	Weak stimulation → pleasure; strong stimulation → punishment-like effects.
Punishment Centers	Location	Central gray area of mesencephalon, periventricular zones, amygdala, hippocampus.
	Function	Suppresses reward centers; induces fear, punishment, and escape responses.

Emotional Patterns	Rage Behavior	Triggered by punishment center stimulation: defense postures, hissing, growling, and aggression. Suppressed by ventromedial nuclei, hippocampus, anterior limbic cortex.
	Placidity and Tamelessness	Induced by stimulation of reward centers.
Reward and Punishment in Learning	Influence on Behavior	Reward encourages continued activity; punishment discourages it.
	Role in Learning	Reward/punishment enhances memory; indifferent stimuli lead to habituation.
Effect of Tranquilizers	Function	Chlorpromazine – Suppresses reward and punishment centers, reducing emotional reactivity.

Hippocampus



HIPPOCAMPUS:

- The hippocampus is a vital part of the brain involved in various cognitive functions. It is located in the medial temporal lobe, forming part of the limbic system, which regulate emotions, learning, memory.

FUNCTIONS:

- **INTEGRATION AND RELAY OF SENSORY SIGNALS:** The hippocampus receives sensory input from multiple cortical regions, processes the input, and sends output signals to areas like the anterior thalamus, hypothalamus, other limbic system structures via fornix. It serves as a conduit for sensory signals to trigger behavioral reactions.
- **BEHAVIORAL MODULATION:** Stimulation of hippocampus can elicit a wide range of emotional and behavioral responses, including pleasure, rage, passivity, excessive sexual drive.

Cont.

- **MEMORY FORMATION AND LEARNING:** The hippocampus is critical for forming new memories, particularly episodic and spatial memories. It help consolidate information from short term memory to long term memory. Damage to hippocampus, such as Alzheimer's disease or anoxic brain injury, severely impairs memory formation and spatial navigation.
- **HYPEREXCITABILITY AND SEIZURES:** The hippocampus prone to hyperexcitability , and weak electrical stimuli can lead to focal epileptic seizures that last longer than the stimuli.
- **CONNECTION TO LIMBIC SYSTEM:** The hippocampus interacts closely with the limbic system(amygdala, hypothalamus, septum, and mammillary bodies) playing a role in emotional regulation, memory encoding and retrieval.

ANTEROGRADE AMNESIA AND HIPPOCAMPAL REMOVAL:

- **Effects of Bilateral Hippocampal Removal**

Individuals with bilateral hippocampal removal (e.g., for epilepsy treatment) experience anterograde amnesia, meaning they lose the ability to form new long-term memories.

- **Preserved Abilities:**

- Retain most previously learned memories.
- Can recall events and information learned before the surgery.
- Can perform short-term memory tasks for a few seconds to minutes.

- **Impaired Abilities:**

- Unable to consolidate new verbal or symbolic memories.
- Difficulty learning new names or associating with daily experiences.
- Example: Such individuals may repeatedly meet the same person without recalling the prior interaction.

Alzheimer's symptoms



memory loss



Problems with words



Lack of temporal and spatial orientation



Problems with daily routine

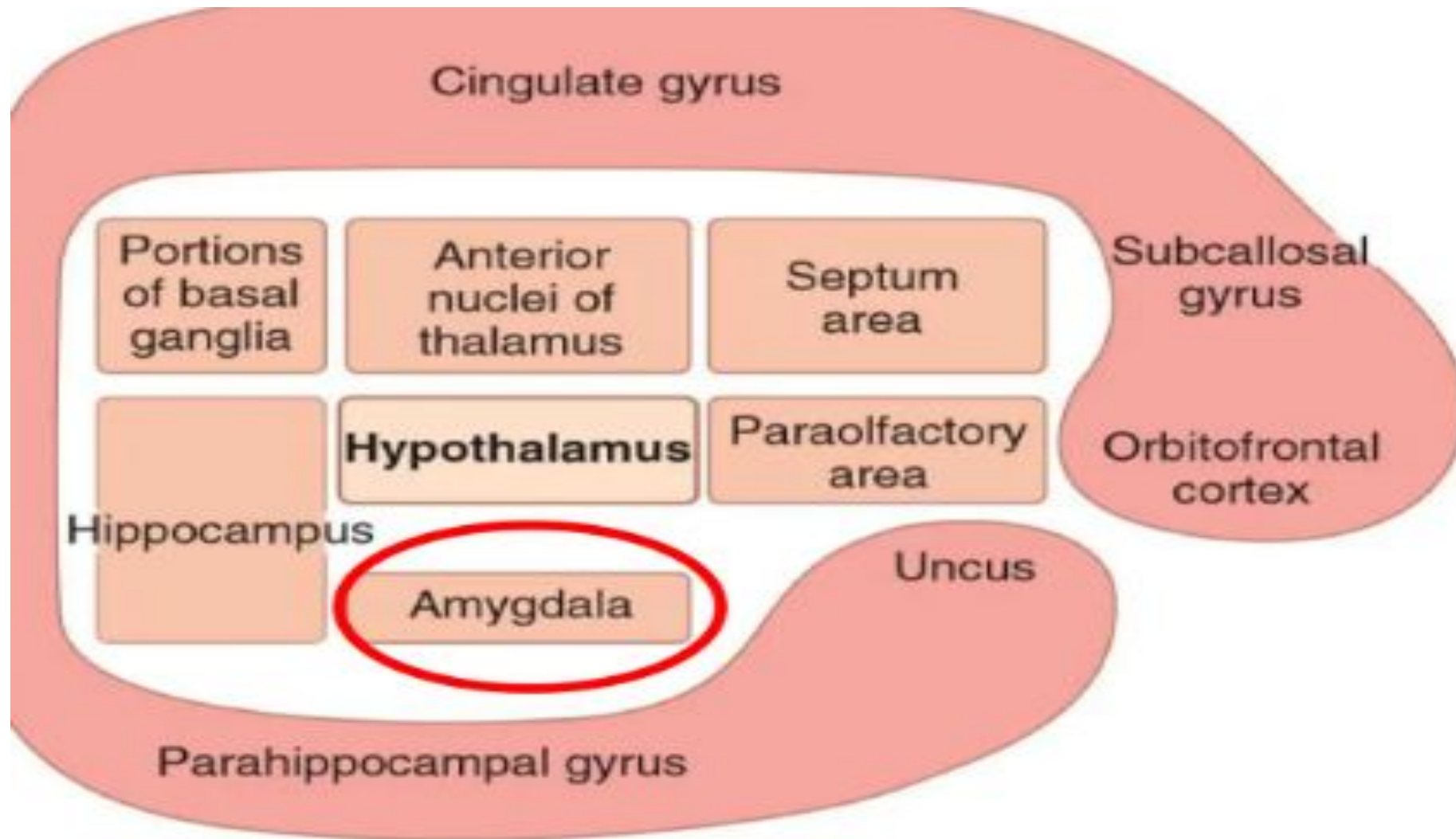


Emotional disturbance



Problems maintaining attention during a discussion

AMYGDALA:



AMYGDALA:

- The amygdala plays a significant role in **emotional processing, behavior, and various autonomic functions.**

Located **beneath the cerebral cortex** in the anterior pole of the temporal lobe. Has strong **connections with the hypothalamus, limbic system, and neocortex (temporal, parietal, and occipital lobes).**

Functions:

- **Olfactory Processing in Lower Animals:**
 - In lower animals, the amygdala is involved in **processing olfactory (smell) stimuli.**
- **Behavioral and Emotional Processing in Humans**
 - In humans, the amygdala is more developed in the **basolateral nuclei** and is associated with **various behaviors, emotions, and motivational activities.**
- **Integrating information from sensory areas :**
 - The amygdala acts as a "window" through which the limbic system perceives the person's relationship to the world by **integrating information from sensory areas (e.g., visual, auditory).**

Regulation of Autonomic Functions

Through its connections with the **hypothalamus**, the amygdala influences:

- **Heart rate and blood pressure**
- **Gastrointestinal motility and secretion**
 - **Defecation, micturition**
- **Pupillary responses (dilation/constriction)**
 - **Secretion of pituitary hormones like gonadotropins and ACTH**

EFFECTS OF STIMULATION

Autonomic Responses:

Stimulation can cause changes in heart rate, blood pressure, gastrointestinal activity, and other physiological responses

Involuntary Movements:

Includes tonic movements (e.g., head raising, body bending), circling, and rhythmic movements related to eating behaviors (licking, chewing, swallowing)

Emotional Responses :

Can induce feelings of rage, fear, and aggression, or elicit reward and pleasure

Sexual Behavior:

Stimulation can lead to sexual activities like erection, copulation, and uterine activity (ovulation, labor)

EFFECTS OF BILATERAL AMYGDALA REMOVAL (KLÜVER-BUCY SYNDROME):

Loss of Fear

The individual or animal shows **no fear of threats**

Extreme Curiosity

Affected animals may demonstrate **excessive curiosity and an impulse to explore everything**

Memory Issues

Rapid forgetting and loss of memory

Inappropriate Behavior

Includes **placing objects in the mouth (even non-food items), and hypersexual behaviors such as attempting copulation with inappropriate animals or objects**

KLÜVER-BUCY SYNDROME:

Bilateral Temporal Lobe & Amygdala Dysfunction

ETIOLOGY & PATHOGENESIS (The Injury)



- Bilateral Anterior Temporal Lobe Damage
- Amygdala & Hippocampus Involvement
- Herpes Simplex Encephalitis (most common)
- Trauma, Stroke, Alzheimer's, Pick's Disease
- Loss of emotional processing & memory integration

CLINICAL FEATURES (The Syndrome)



Placidity (Dumbness, loss of fear/anger)



Hyperorality (Oral exploration of objects)



Hypersexuality (Indiscriminate sexual behavior)



Visual Agnosia (Psychic Blindness, failure to recognize)



Hypermetamorphosis (Compulsive attention to stimuli)



Memory Deficits

DIAGNOSTIC STRATEGY (Assessment)

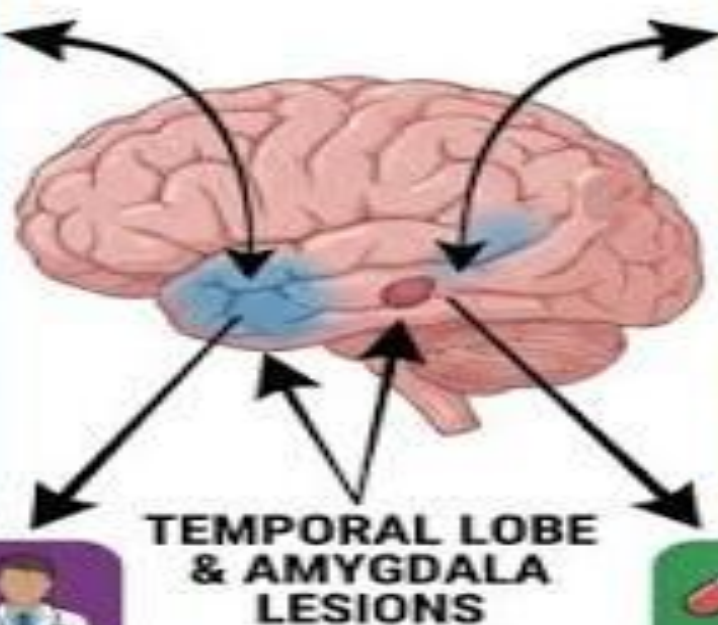


- Clinical History & Exam
- Neuroimaging (MRI/CT) – Temporal lobe atrophy/lesions
- EEG (to rule out seizures)
- Neuropsychological Testing (Visual recognition, memory)
- Viral PCR (if encephalitis suspected)

MANAGEMENT & PROGNOSIS (Care & Outlook)



- No Specific Cure, Supportive Care
- Symptomatic Treatment (e.g., Antipsychotics, SSRIs for behavior)
- Behavioral Therapy & Environmental Modification
- Safety Measures (due to hyperorality)
- Variable Prognosis (Depends on underlying cause)



EL

Korsakoff Syndrome:

- **Cause:**

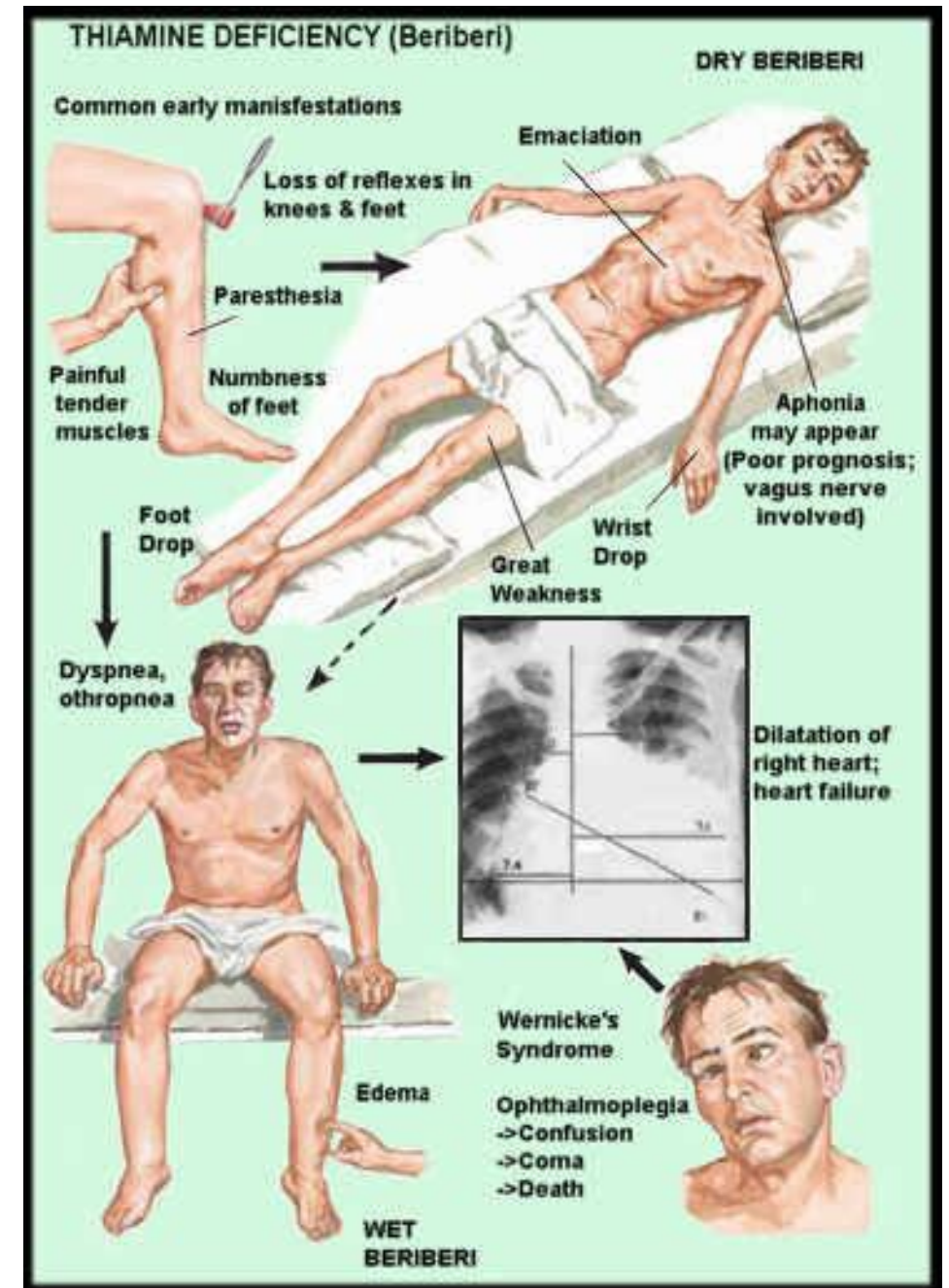
- Alcoholics with thiamine deficiency or Acute presentation of Wernicke encephalopathy.

- **Lesion:**

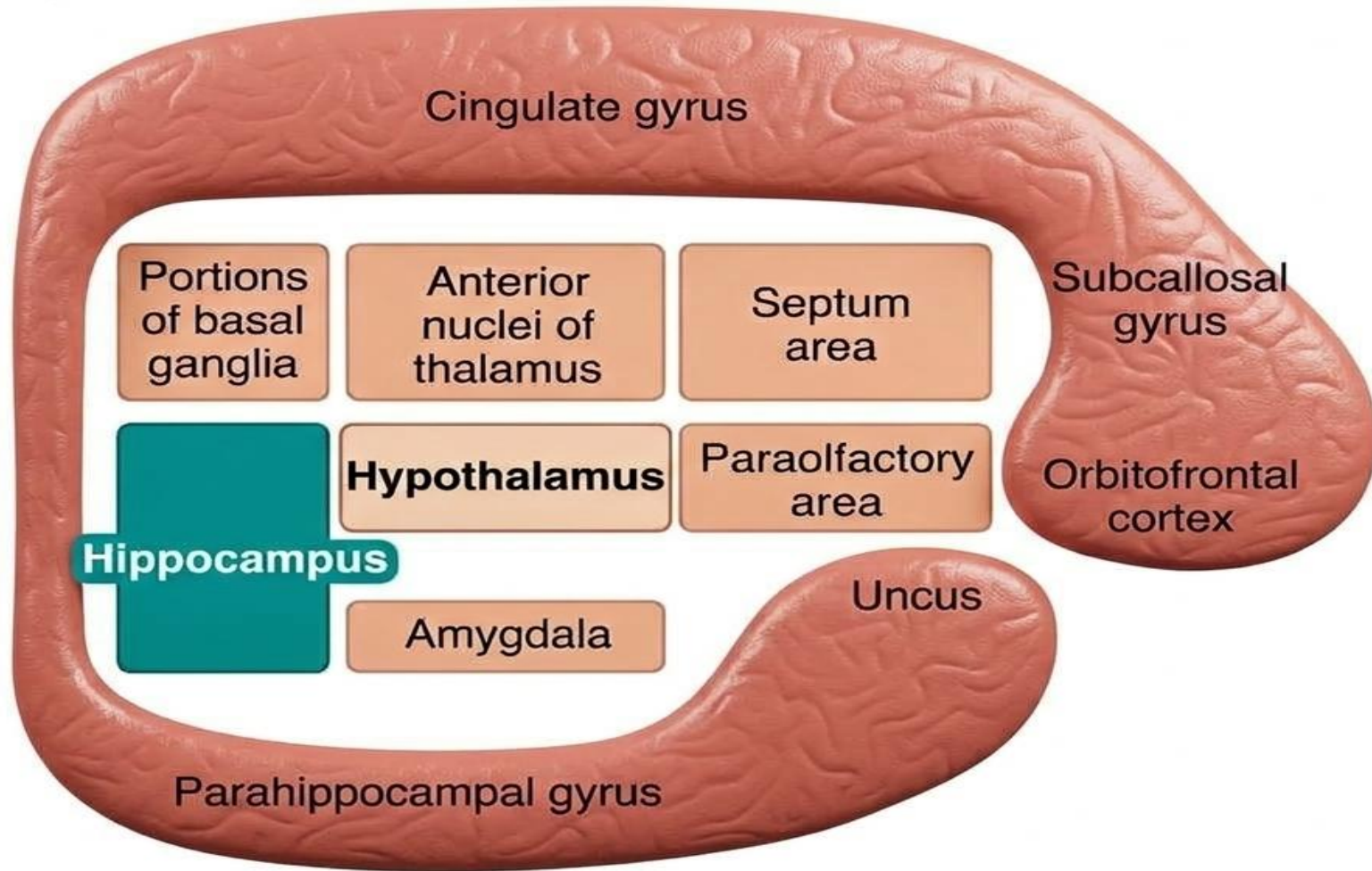
- Lesion in the mammillary bodies and the dorsomedial nuclei of the thalamus

- **Effects:**

- Anterograde and retrograde amnesia
- Confabulation (making up stories to replace past memories they can no longer retrieve)
- Temporospatial disorientation.



LIMBIC CORTEX



LIMBIC CORTEX

The limbic cortex is a part of the limbic system that connects the cerebral cortex and subcortical limbic structures. It acts as a transitional zone for signal transmission between these areas. It is involved in controlling complex behaviors by integrating information from different brain regions.

Role: Serves as an association area that links cortical functions with limbic system functions for behavior control.

STIMULATION AND ABLATION EFFECTS:

- **Anterior Temporal Cortex Ablation:**

- Causes **Klüver-Bucy syndrome** (loss of fear, increased curiosity, inappropriate sexual behavior, excessive object engagement).
- Leads to **insomnia** and **motor restlessness**.

- **Anterior Cingulate and Subcallosal Gyrus Ablation**

- Increases **aggression and rage** by removing prefrontal inhibition of rage centers.

BEHAVIORAL ASSOCIATIONS:

- **Anterior Temporal Cortex**

Involved in **taste and smell-related** behaviors.

- **Parahippocampal Gyri**

Associated with **auditory and complex thought** processes.

- **Cingulate Cortex**

Likely involved in **sensorimotor behavior** coordination

- *Limbic cortex integrates sensory, emotional, & cognitive information to regulate appropriate behavioral response*

Functional Chart of the Limbic System

Part of Limbic System 

~ Functions ~ 

Hippocampus



-  - Involved in memory formation and learning.
-  - Converts short-term memories to long-term memories.
- Hyperexcitability can cause seizures and hallucinations. 
- Removal leads to anterograde amnesia, impairing long-term memory formation. 

 Amygdala

-    - Regulates emotions like fear, rage, and pleasure.   
-  - Mediates physiological responses (e.g., heart rate, blood pressure, GI motility).
-  - Controls sexual behavior and involuntary movements. 
-  - Stimulation triggers rage, fear, and pleasure responses. 
-  - Bilateral ablation causes Klüver-Bucy syndrome 
(loss of fear, excessive curiosity, inappropriate sexual behavior)

Limbic Cortex

- Acts as a transitional zone between the cerebral cortex & subcortical structures.
-  - Influences behavior and sensory processing. 
-  - Destruction of specific areas causes behavior changes (e.g., Klüver-Bucy syndrome, aggression).  
-  - Associated with gustatory, olfactory, auditory, and sensorimotor behavior. 

THANK YOU